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exoTEDRF: An EXOplanet Transit and Eclipse Data Reduction Framework

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Software

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Summary

Since the start of science operations in July 2022, JWST has delivered groundbreaking results on a regular basis. For spectroscopic studies of exoplanet atmospheres the process of extracting robust and reliable atmosphere spectra from the raw JWST observations is critical. Especially as the field pushes to detect the signatures of secondary atmospheres on rocky Earth-like planets it is imperative to ensure that the spectral features that drive atmosphere inferences are robust against the particular choices made during the data reduction process.

Here, I present the community with `exoTEDRF` (EXOplanet Transit and Eclipse Data Reduction Framework; formerly known as `supreme-SPOON`), an end-to-end pipeline for data reduction and light curve analysis of time series observations (TSOs) of transiting exoplanets with JWST. The pipeline is highly modular and designed to produce reliable spectra from raw JWST exposures. `exoTEDRF` consists of four stages, each of which are further subdivided into a series of steps. These steps can either be run individually, for example in a Jupyter notebook, or via the command line using the provided configuration files. The steps are highly tunable, allowing full control over every parameter in the reduction. Each step also produces diagnostic plots to allow the user to verify their results at each intermediate stage, and compare outputs with other pipelines if so desired. Finally, `exoTEDRF` has also been designed to be run in "batch" mode: simultaneously running multiple reductions, each tweaking a subset of parameters, to understand any impacts on the resulting atmosphere spectrum.

Overview of exoTEDRF Stages

Like similar pipelines (`Eureka!` (Bell et al., 2022), `jwst` (Bushouse et al., 2022), etc.) `exoTEDRF` is divided up into four major stages which are summarized below:

- Stage 1, Detector-level processing: Converts raw, 4D (integrations, groups, x-pixel, y-pixel) data frames to 3D (integrations, x-pixel, y-pixel) slope images. Steps include superbias subtractions, correction of $1/f$ noise, ramp fitting, etc.
- Stage 2, Spectroscopic processing: Performs additional calibrations to prepare slope images for spectral extraction. Steps include, flat field correction, background subtraction, etc.
- Stage 3, Spectral extraction: Extract the 2D stellar spectra from the 3D slope images.
- Stage 4, Light curve fitting: An optional stage for the fitting of extracted light curves.

In `exoTEDRF`, Stage 4 is an optional installation which is currently built around the excellent `juliet` library (Néstor Espinoza et al., 2019), and incorporates tools such as `ExoTiC-LD` (Grant & Wakeford, 2022) for the estimation of stellar limb darkening parameters. In certain places (e.g., superbias subtraction, flat field correction), `exoTEDRF` simply provides a wrapper around the existing functionalities of the `jwst` package maintained by the Space Telescope Science Institute.

Statement of Need

Data analysis is a challenging process that is encountered by all observational studies. Ensuring that the resulting atmosphere spectra are robust against particular choices made in the reduction process is critical, especially as we push to characterize the atmospheres of small rocky planets. The modularity and tunability of `exoTEDRF` make it easy to run multiple reductions of a given dataset, and therefore robustly ascertain whether the spectral features driving atmosphere inferences are robust, or sensitive to the peculiarities of a given reduction. Moreover, `exoTEDRF` has full support for TSOs with NIRISS/SOSS (Albert et al., 2023), an observing mode which is underserved by the current ecosystem of JWST reduction tools, including the ability to run the `ATOCA` extraction algorithm (Darveau-Bernier et al., 2022; Radica et al., 2022) to explicitly model the SOSS order overlap.

Documentation

Documentation for `exoTEDRF`, including example notebooks, is available at https://exotedrf.readthedocs.io/en/latest/.

Uses of exoTEDRF in Current Literature

`exoTEDRF` (particulalry in its previous life as `supreme-SPOON`) has been widely applied to exoplanet TSOs. A list of current literature which has made use of `exoTEDRF` includes: A. D. Feinstein et al. (2023), Coulombe et al. (2023), Radica et al. (2023), Albert et al. (2023), Lim et al. (2023), Radica et al. (2024), Fournier-Tondreau et al. (2024), and Benneke et al. (2024).

Future Developments

The current release of `exoTEDRF` (v1.4.0) currently supports the reduction of TSOs observed with JWST NIRISS/SOSS. Support for observations with NIRSpec and MIRI/LRS are in development and will be added in the coming months. `exoTEDRF` has also been applied to exoplanet observations from the Hubble Space Telescope using the UVIS mode (Radica et al., 2024, in prep). This functionality will also be made available to the public. Finally, updates to the light curve fitting routines are underway to allow for more flexibility for the fitting of both astrophysical and systematics models.

Suggestions for additional features are always welcome!

Similar Tools

The following is a list of other open source pipelines tailored to exoplanet observations with JWST: `Eureka!` (Bell et al., 2022), `jwst` (Bushouse et al., 2022), `tshirt` (Schlawin & Glidic, 2022), `PACMAN` (Zieba & Kreidberg, 2022), `nirHiss` (A. Feinstein, 2022), `ExoTiC-JEDI` (Alderson et al., 2022), `ExoTiC-MIRI` (Grant et al., 2023), and `transitspectroscopy` (Nestor Espinoza, 2022). Packages like `exoplanet` (Foreman-Mackey et al., 2021) and `Eureka!` (Bell et al., 2022) also enable similar light curve fitting.

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The foundations of `exoTEDRF` are built upon many wonderful Python libraries, including `numpy` (Harris et al., 2020), `scipy` (Virtanen et al., 2020), `astropy` (Astropy Collaboration et al., 2013, 2018), and `matplotlib` (Hunter, 2007).

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